

STATEMENT OF WORK – Purchase of optomechanical supplies for Solar Cell Characterization

1. Objective/Requirements

NASA seeks to procure a range of optomechanical supplies and detectors to aid in the development of a custom solar cell photoluminescence and carrier lifetime characterization system. The supplies include optical filters/mirror with high damage threshold, mechanical equipment for mounting optics, photon detectors, and translation stages.

2. Characteristics, Scope, and Specifications

- Custom characterization setup utilizes a 10W CW 532nm laser and 3.5W 400-1700nm tunable Supercontinuum laser (50ps pulse widths)
 - Optical equipment shall have damage thresholds for lasers provided above. And includes:
 - Microscope objective, Infinity Corrected, 400 – 1100nm ARC, 0.75NA, 50X magnification.
 - Tube lens for objective.
 - Including filter set for shortpass and longpass filtration (SM1 compatible)
 - Neutral Density Filters (OD: 3&6) for 350nm-700nm and 650nm – 1050nm
 - 600nm -1700nm beam splitter plates
 - Dichroic mirrors for 650nm – 1600nm transmission (400 -633nm reflection) and 825 – 1300nm transmission (400-785nm reflection)
 - Adjustable Zoom beam expander (400nm – 650nm) & (650nm – 1050nm)
 - Zoom fiber collimator,
 - Detectors include:
 - Power & Energy Meter
 - Si Sensor for 500 nW – 500mW measurements
 - Thermopile Sensor for 100uW – 5W measurements
 - Compact Si Sensor (with SMA attachment) for 20nW – 50mW measurements
 - 400nm -1000nm Single Photon detector capable of measuring lifetimes from 1ns (or lower) to 1ms
 - Mechanical supplies shall include:
 - Filter wheels stations to hold filter sets and Neutral Density Filters. Holds max capacity of 12 filters.
 - Post holders, optical posts, and mounting posts, optical rails, table clamps and breadboards.
 - X-Y translation stage, 150 mm travel range
 - Z translation stage, 50mm travel range
 - Controllers for translation stages
 - Flip mounts
 - SM1 tube Lens covers

- Laser safety equipment include:
 - 4 Hinged L-Shaped optical enclosure , 21" x 15" x 12" (L x W x H)
capable of being joined together to form one enclosure